

Abba Otieno Ndomo

Atlanta, GA | 404-819-5945 | abba.ndomo@gmail.com | Portfolio: <https://andomo3.github.io/baah>
Work Authorization: F-1 | CPT available now | OPT (May 2027) | STEM OPT eligible

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science, Industrial & Systems Engineering — Operations Research

Atlanta, GA
Aug. 2023 – Dec 2027

- **Honors & Awards:** Dean's List, Peer Inspiration Award @ ISyE x CASE, 3rd Place @ OKSEF International Engineering Fair
- **Coursework:** Machine Learning, Stochastic Processes, Linear & Non-Linear Optimization, Database Systems, Data Structures & Algorithms, Simulations
- **Major GPA:** 3.41

PROFESSIONAL EXPERIENCE

ML Intern - Altura

Python, PySpark, Airflow, pandas, PyArrow, DuckDB, Next.js, Parquet

Jan. 2026 – Present
Atlanta, GA

- Built a computer-vision measurement pipeline that estimates cabinet linear footage and countertop area from real-estate listing photos, chaining CLIP zero-shot filtering, Grounding DINO open-vocabulary detection, and SAM seam segmentation, with a Claude Vision fallback when no scale reference is recoverable.
- Diagnosed scale-anchor selection — not detection quality — as the dominant term in the measurement error budget, then calibrated a 0.45 detection-confidence floor from 8 labeled anchor observations, cutting mean cabinet-width error from 44.5% to 18.2% on the like-for-like validation set.
- Enforced a strict layered architecture — agents orchestrate, modules hold business logic, typed Pydantic models carry all data across boundaries — and lazy-imported the heavy ML dependencies so the full 289-test suite runs in ~42 seconds with no network, model weights, or API keys.

SWE Intern – KPMG

Python, PostgreSQL, Statistical Analysis, System Design

Jun. 2024 –Jan. 2025
Mumbai, India

- Architected a fault-tolerant 3-tier data processing pipeline ingesting, transforming, and classifying 10,000+ monthly PDFs into a structured PostgreSQL data store; served 50+ concurrent auditors across 3 continents with 95% classification precision (held-out test, p less than 0.001) and zero schema violations across 120,000+ processed documents
- Reduced end-to-end pipeline latency 45s → 8s (82%) through algorithmic optimization ($O(n^2) \rightarrow O(n+m)$), parallel batch processing via Python multiprocessing, and database optimization (composite indexes, connection pooling, query restructuring)
- Engineered a data quality framework with 50+ automated validation checks — distributional tests, Z-score outlier detection, referential integrity, completeness metrics — catching schema drift and data anomalies pre-transformation; reduced downstream data errors from 40% → 5% and prevented 12+ cascade failures across dependent analytics pipelines.
- Achieved 99% test coverage with CI/CD pipeline validating all data transformations on every deployment; system processed 120,000+ documents over 14 months with zero production incidents and \$62K annual cost savings vs. manual processing

Quantitative Analyst - GTSF

Python, Bloomberg Terminal API (blpapi), pandas, Statsmodels, NumPy, ARIMA, Monte Carlo

Jan. 2025 – May 2025
Atlanta, GA

- Built automated data ingestion system integrating Bloomberg Terminal API across 120 equities (2.4x coverage expansion); engineered concurrent request batching with exponential-backoff retry logic handling rate limits, stale tickers, and corporate action events — sustaining sub-minute data freshness across a \$500K live equity portfolio.
- Engineered time-series analytics pipeline processing historical price data through ARIMA models and rolling correlation matrices; ran 10,000-iteration Monte Carlo simulations using historical covariance structure across 120 equities — identifying 8% portfolio downside risk under macro shock scenarios and reducing VaR by 15% through data-driven hedging decisions.

PROJECTS

Founder - PerChance

PySpark, Airflow, PyTorch, XGBoost, Django, Postgres, Docker, Redis, React

Jan. 2026 – Present

- Architected an end-to-end data platform serving 500+ daily active users with 500,000+ daily API calls; automated data ingestion, transformation, feature engineering, model training, and prediction serving via orchestrated Airflow DAGs — eliminating manual pipeline intervention across the full ML lifecycle.
- Built scalable PySpark ETL pipeline processing 500,000+ NBA statistical records with custom Holt-Winters time-decay feature engineering; Airflow orchestration of full ETL → feature store → model retraining → serving pipeline reduced ML prep time 90% (from 2 hours to 12 minutes)
- Designed high-throughput 3-tier data architecture (Django API / PostgreSQL database / React frontend) with Redis caching layer and Celery async workers; composite indexes on high-cardinality columns and query optimization sustain sub-200ms p95 API latency under live production traffic
- Implemented production observability with Prometheus/Grafana tracking p50/p95/p99 request latencies, error rates, and pipeline health; configured alerting thresholds and automated rollback triggers — maintaining 99.5% uptime over 6 months of continuous operation with zero production incidents.
- Engineered a gradient-boosted regression stack (XGBoost, CatBoost, scikit-learn) with SHAP attribution and strictly chronological train/test splits, feeding five analysis services: edge calibration, floor/ceiling distribution profiling, per-opponent exploitability, behavioral fingerprinting, and variance decomposition.

Credit Classification System

Python, FastAPI, XGBoost, PostgreSQL, Redis, Airflow, Docker, Prometheus, Grafana, dbt, Great Expectations, MLflow

Aug. 2025 – Dec. 2025

- Architected an end-to-end credit risk prediction system serving real-time loan decisions via FastAPI REST API; Redis caching on feature hash keys sustains sub-10ms p95 inference latency with 70%+ cache hit rate, and Pydantic v2 input validation prevents malformed applications from reaching the model layer
- Engineered a full MLOps pipeline with Airflow orchestrating weekly ingestion → dbt feature transformations → Great Expectations validation → XGBoost retraining → MLflow model registry promotion; new models deploy only when AUC exceeds the current production model on a held-out evaluation set
- Implemented KS-test-based prediction drift detection monitoring the output probability distribution against a rolling baseline; alerts trigger automated retraining when p-value drops below 0.05, preventing silent model degradation under distributional shift

LEADERSHIP

Housing Office Assistant, Lead Conference Assistant

Georgia Institute of Technology Housing Department

Sept.2025 – Present
Atlanta, GA

- Led Delivery & Mail Operations Team; Developed Standard Operating Procedures to coordinate 50+ monthly events serving 1000+ students,

TECHNICAL EXPERTISE

Languages: Python (OOP, concurrency, async), SQL (CTEs, Window Functions, query optimization), Java, JavaScript, TypeScript, Go (learning), Bash

Systems & Data Engineering: Airflow, PostgreSQL, DuckDB, Parquet, PyArrow, ETL/ELT Pipelines, Data Modeling (3NF, Snowflake), Data Warehousing, EERD

Machine Learning: PyTorch, XGBoost, Scikit-Learn, SHAP, Feature Engineering, A/B Testing, Experiment Design, RAG Pipelines, NLP (Transformers)